

# Data Analysis with Python

## Cheat Sheet: Exploratory Data Analysis

| Package/Method                 | Description                                                                                                                                        | Code Example                                                                                                                                             |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Complete dataframe correlation | Correlation matrix created using all the attributes of the dataset.                                                                                | <pre>df.corr()</pre>                                                                                                                                     |
| Specific Attribute correlation | Correlation matrix created using specific attributes of the dataset.                                                                               | <pre>df[['attribute1','attribute2',...]].corr()</pre>                                                                                                    |
| Scatter Plot                   | Create a scatter plot using the data points of the dependent variable along the x-axis and the independent variable along the y-axis.              | <pre>from matplotlib import pyplot as plt plt.scatter(df[['attribute_1']],df[['attribute_2']])</pre>                                                     |
| Regression Plot                | Uses the dependent and independent variables in a Pandas data frame to create a scatter plot with a generated linear regression line for the data. | <pre>import seaborn as sns sns.regplot(x='attribute_1',y='attribute_2', data=df)</pre>                                                                   |
| Box plot                       | Create a box-and-whisker plot that uses the pandas dataframe, the dependent, and the independent variables.                                        | <pre>import seaborn as sns sns.boxplot(x='attribute_1',y='attribute_2', data=df)</pre>                                                                   |
| Grouping by attributes         | Create a group of different attributes of a dataset to create a subset of the data.                                                                | <pre>df_group = df[['attribute_1','attribute_2',...]]</pre>                                                                                              |
| GroupBy statements             | a. Group the data by different categories of an attribute, displaying the average value of numerical attributes with the same category.            | <pre>a) df_group = df.groupby(['attribute_1'],as_index=False).mean() b) df_group = df.groupby(['attribute_1','attribute_2'],as_index=False).mean()</pre> |

|                                 |                                                                                                                                                |                                                                                                                   |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
|                                 | b. Group the data by different categories of multiple attributes, displaying the average value of numerical attributes with the same category. |                                                                                                                   |
| Pivot Tables                    | Create Pivot tables for better representation of data based on parameters                                                                      | <code>grouped_pivot = df_group.pivot(index='attribute_1',columns='attribute_2')</code>                            |
| Pseudocolor plot                | Create a heatmap image using a PsuedoColor plot (or pcolor) using the pivot table as data.                                                     | <code>from matplotlib import pyplot as plt<br/>plt.pcolor(grouped_pivot, cmap='RdBu')</code>                      |
| Pearson Coefficient and p-value | Calculate the Pearson Coefficient and p-value of a pair of attributes                                                                          | <code>From scipy import stats<br/>pearson_coef,p_value=stats.pearsonr(df['attribute_1'],df['attribute_2'])</code> |

